🧠 **C# Collections – Real-Time Assignments**

📘 **1. List – *E-Commerce Product Catalog***

🎯 **Scenario:**

You are building a product listing page.

📝 **Assignment:**

1. Create a Product class with:
    - Id, Name, Price, Category
2. Store multiple products using List<Product>
3. Perform the following:
    - Add 10 products
    - Display all products
    - Find products with price > 1000
    - Sort products by price (ascending & descending)
    - Remove a product by Id

💡 **Bonus:**

- Use LINQ to filter products by category

📗 **2. Dictionary<TKey, TValue> – *Student Management System***

🎯 **Scenario:**

Store student data with unique IDs.

📝 **Assignment:**

1. Create a Student class:
    - Id, Name, Marks

2. Use Dictionary<int, Student>
3. Perform:
    - Add 5 students
    - Retrieve student by Id
    - Check if a student exists using ContainsKey
    - Update marks of a student
    - Remove a student

💡 **Bonus:**

- Print all students who scored above 75

📕 **3. HashSet – *Online Event Registration***

🎯 **Scenario:**

Ensure no duplicate registrations.

📝 **Assignment:**

1. Use HashSet<string> for storing email IDs
2. Perform:
    - Add 10 emails (include duplicates intentionally)
    - Display unique emails
    - Check if a specific email is registered
    - Remove an email

💡 **Bonus:**

- Compare two event lists and find common participants

📕 **4. Stack – *Undo Feature in Text Editor***

🎯 **Scenario:**

Implement undo functionality.

📝 **Assignment:**

1. Use Stack<string> to store actions
2. Perform:
    - Push actions (e.g., "Type A", "Delete B")
    - Undo last 3 actions using Pop()
    - Display current top action using Peek()

💡 **Bonus:**

- Implement redo using another stack

📚 **5. Queue – *Hospital Patient Queue***

🎯 **Scenario:**

Patients are treated in FIFO order.

📝 **Assignment:**

1. Create a Patient class:
    - Id, Name, Disease
2. Use Queue<Patient>
3. Perform:
    - Add 5 patients
    - Serve (Dequeue) 2 patients
    - Display next patient
    - Show all remaining patients

💡 **Bonus:**

- Add priority handling (VIP patients)

🧮 6. LinkedList – *Music Playlist System*

🎯 **Scenario:**

Songs can be added anywhere in playlist.

📝 **Assignment:**

1. Create a Song class:
    - Id, Title, Artist
2. Use LinkedList<Song>
3. Perform:
    - Add songs at beginning, end, and middle
    - Remove a specific song
    - Traverse forward and backward
    - Find a song by title

💡 **Bonus:**

- Implement "Play Next" feature